

GNU nano 7.2

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <time.h>
#include <string.h>

int main() {
    int fd[2];
    pid_t pid;
    char buffer[100];

    if (pipe(fd) == -1) {
        perror("pipe");
        return 1;
    }

    pid = fork();

    if (pid < 0) {
        perror("fork");
        return 1;
    }

    if (pid > 0) {
        // Parent - Producer
        close(fd[0]);

        clock_t start = clock();

        char *message = "Hello from Producer";

        write(fd[1], message, strlen(message) + 1);

        clock_t end = clock();

        double time_taken =
            ((double)(end - start)) / CLOCKS_PER_SEC;
    }
}
```

```

    printf("Producer: Data sent: %s\n", message);
    printf("Communication time: %f seconds\n", time_taken);

    close(fd[1]);

    wait(NULL);
} else {
    // Child - Consumer
    close(fd[1]);

    read(fd[0], buffer, sizeof(buffer));

    printf("Consumer: Data received: %s\n", buffer);

    close(fd[0]);
}

return 0;
}|

```

```

sravya@Sravya1955:~/os$ nano os1.1.7.c
sravya@Sravya1955:~/os$ gcc os1.1.7.c -o os
sravya@Sravya1955:~/os$ ./os
Producer: Data sent: Hello from Producer
Consumer: Data received: Hello from Producer
Communication time: 0.000085 seconds

```

```
sravya@Sravya1955:~/os$ ls -l | grep ".c"
```

```
\-rw-r--r-- 1 sravya sravya 927 Aug 27 09:28 1.1.1.c
-rw-r--r-- 1 sravya sravya 676 Aug 27 09:07 1.c
-rw-r--r-- 1 sravya sravya 927 Aug 27 09:22 2.c
-rw-r--r-- 1 sravya sravya 927 Aug 27 09:33 3.c
-rw-r--r-- 1 sravya sravya 1 Aug 27 09:34 4.c
drwxr-xr-x 2 sravya sravya 4096 Aug 26 03:15 FileDescriptores
drwxr-xr-x 8 sravya sravya 4096 Aug 1 03:46 MyProject
drwxr-xr-x 2 sravya sravya 4096 Aug 7 13:28 docs
drwxr-xr-x 2 sravya sravya 4096 Jul 31 09:34 exec
-rwxr-xr-x 1 sravya sravya 16224 Aug 5 03:48 filecopy
-rw-r--r-- 1 sravya sravya 681 Aug 5 03:32 filecopy.c
-rw-r--r-- 1 sravya sravya 93 Jul 29 03:11 os.c
-rw-r--r-- 1 sravya sravya 207 Aug 5 04:00 os1.1.1.c
-rw-r--r-- 1 sravya sravya 422 Aug 5 04:01 os1.1.2.c
-rw-r--r-- 1 sravya sravya 169 Aug 5 04:03 os1.1.3.c
-rw----- 1 sravya sravya 0 Aug 5 04:05 os1.1.4.c.save
-rw-r--r-- 1 sravya sravya 708 Aug 5 04:10 os1.1.5.c
-rw-r--r-- 1 sravya sravya 507 Aug 7 09:03 os1.1.6.c
-rw-r--r-- 1 sravya sravya 1025 Aug 28 08:12 os1.1.7.c
-rw-r--r-- 1 sravya sravya 872 Aug 5 03:22 os1.1.c
-rw-r--r-- 1 sravya sravya 713 Aug 5 03:27 os1.2.c
-rw-r--r-- 1 sravya sravya 680 Aug 5 03:34 os1.3.c
-rw-r--r-- 1 sravya sravya 508 Aug 5 03:53 os1.4.c
-rw-r--r-- 1 sravya sravya 89 Jul 29 03:17 os1.c
-rw-r--r-- 1 sravya sravya 0 Jul 29 03:16 os1.xc
-rw-r--r-- 1 sravya sravya 203 Aug 1 02:49 os14.c
-rw-r--r-- 1 sravya sravya 243 Aug 1 02:50 os15.c
-rw-r--r-- 1 sravya sravya 154 Aug 11 06:33 os2.c
-rw-r--r-- 1 sravya sravya 279 Jul 29 03:23 os3.c
-rw-r--r-- 1 sravya sravya 214 Jul 29 03:26 os4.c
-rw-r--r-- 1 sravya sravya 116 Jul 29 03:26 os5.c
-rw-r--r-- 1 sravya sravya 220 Jul 29 03:29 os6.c
-rw-r--r-- 1 sravya sravya 132 Jul 29 03:30 os7.c
-rw-r--r-- 1 sravya sravya 623 Jul 29 03:38 os8.c
-rw-r--r-- 1 sravya sravya 977 Aug 11 06:59 parser.c
-rw-r--r-- 1 sravya sravya 98 Aug 25 06:47 program1.c
drwxr-xr-x 2 sravya sravya 4096 Aug 7 09:36 project
-rw-r--r-- 1 sravya sravya 756 Aug 11 07:03 quotes.c
-rw-r--r-- 1 sravya sravya 423 Aug 11 06:56 shell.c
-rw-r--r-- 1 sravya sravya 1308 Aug 11 06:57 shell_history.c
```

```
-rw-r--r-- 1 sravya sravya 1308 Aug 11 08:00 shell_history.c.save
-rw-r--r-- 1 sravya sravya 70 Aug 5 03:30 source.txt
drwxr-xr-x 2 sravya sravya 4096 Aug 7 13:28 src
-rw-r--r-- 1 sravya sravya 96 Aug 25 06:45 std.c
-rw-r--r-- 1 sravya sravya 97 Aug 25 06:47 std.c.save
```

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    int fd[2];
    pid_t p1, p2;

    if (pipe(fd) == -1) {
        perror("pipe");
        return 1;
    }

    // First child - executes ls -l
    p1 = fork();

    if (p1 < 0) {
        perror("fork");
        return 1;
    }

    if (p1 == 0) {
        // Redirect stdout to pipe
        dup2(fd[1], STDOUT_FILENO);

        close(fd[0]);
        close(fd[1]);

        execlp("ls", "ls", "-l", NULL);

        perror("execlp ls");
        return 1;
    }

    // Second child - executes grep ".c"
    p2 = fork();
```

```
if (p2 < 0) {
    perror("fork");
    return 1;
}

if (p2 == 0) {
    // Redirect stdin from pipe
    dup2(fd[0], STDIN_FILENO);

    close(fd[0]);
    close(fd[1]);

    execlp("grep", "grep", ".c", NULL);

    perror("execlp grep");
    return 1;
}

// Parent closes both ends
close(fd[0]);
close(fd[1]);

// Wait for both children
waitpid(p1, NULL, 0);
waitpid(p2, NULL, 0);

return 0;
}
```

```
sravya@Sravya1955:~/os$ gcc os1.1.8.c -o os
```

```
sravya@Sravya1955:~/os$ ./os
```

```
-rw-r--r-- 1 sravya sravya 927 Aug 27 09:28 1.1.1c
-rw-r--r-- 1 sravya sravya 676 Aug 27 09:07 1.c
-rw-r--r-- 1 sravya sravya 927 Aug 27 09:22 2.c
-rw-r--r-- 1 sravya sravya 927 Aug 27 09:33 3.c
-rw-r--r-- 1 sravya sravya 1 Aug 27 09:34 4.c
drwxr-xr-x 2 sravya sravya 4096 Aug 26 03:15 FileDescriptors
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-rw-r--r-- 1 sravya sravya 169 Aug 5 04:03 os1.1.3.c
-rw----- 1 sravya sravya 0 Aug 5 04:05 os1.1.4.c.save
-rw-r--r-- 1 sravya sravya 708 Aug 5 04:10 os1.1.5.c
-rw-r--r-- 1 sravya sravya 507 Aug 7 09:03 os1.1.6.c
-rw-r--r-- 1 sravya sravya 1025 Aug 28 08:12 os1.1.7.c
-rw-r--r-- 1 sravya sravya 1066 Aug 28 08:18 os1.1.8.c
-rw-r--r-- 1 sravya sravya 872 Aug 5 03:22 os1.1.c
-rw-r--r-- 1 sravya sravya 713 Aug 5 03:27 os1.2.c
-rw-r--r-- 1 sravya sravya 680 Aug 5 03:34 os1.3.c
-rw-r--r-- 1 sravya sravya 508 Aug 5 03:53 os1.4.c
-rw-r--r-- 1 sravya sravya 89 Jul 29 03:17 os1.c
-rw-r--r-- 1 sravya sravya 0 Jul 29 03:16 os1.xc
-rw-r--r-- 1 sravya sravya 203 Aug 1 02:49 os14.c
-rw-r--r-- 1 sravya sravya 243 Aug 1 02:50 os15.c
-rw-r--r-- 1 sravya sravya 154 Aug 11 06:33 os2.c
-rw-r--r-- 1 sravya sravya 279 Jul 29 03:23 os3.c
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drwxr-xr-x 2 sravya sravya 4096 Aug 7 09:36 project
-rw-r--r-- 1 sravya sravya 756 Aug 11 07:03 quotes.c
```

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-rw-r--r-- 1 sravya sravya 423 Aug 11 06:56 shell.c
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-rw-r--r-- 1 sravya sravya 97 Aug 25 06:47 std.c.save
sravya@Sravya1955:~/os$ |
```